Differential Calculus I

Chapter 2

2.1 INTRODUCTION

Differential calculus is the study of derivative, i.e., the study of change of functions w.r.t. the change in inputs. It is the mathematical study of change, motion, growth or decay, etc. In this chapter, we will study successive differentiation, mean value theorems, such as Rolle's theorem, Lagrange's mean value theorem, Cauchy' mean value theorem, expansion of functions and indeterminate forms.

2.2 SUCCESSIVE DIFFERENTIATION

If $y = f(x)$ be a differentiable function of x, then its derivative $\frac{dy}{dx}$ is called the first order derivative of y and is in general a function of x. If $\frac{dy}{dx}$ is differentiable, then its derivative is called the second order derivative of y and is denoted by $\frac{d^2y}{dx^2}$. Similarly, the derivative of $\frac{d^2y}{dx^2}$ is called the third order derivative of y and is denoted by $\frac{d^3y}{dx^3}$ and so on.

The successive differential coefficients of the function $y = f(x)$ are denoted by

$$\frac{dy}{dx}, \frac{d^2y}{dx^2}, \ldots, \frac{d^ny}{dx^n}, \ldots$$

Alternative methods of writing the differential coefficients are

where $D = \frac{d}{dx}$

$Dy, D^2y, D^3y, \ldots\ldots, D^ny, \ldots$

$f'(x), f''(x), f'''(x), \ldots, f^n(x), \ldots$

$y'(x), y''(x), y'''(x), \ldots, y^n(x), \ldots$

$y_1(x), y_2(x), y_3(x), \ldots, y_n(x), \ldots$

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The value of nth differential coefficient at $x = a$ is denoted by

$$\left(\frac{d^n y}{dx^n}\right)_{x=a} \quad \text{or} \quad (y_n)_a \quad \text{or} \quad f^n(a) \quad \text{or} \quad y^n(a).$$

2.2.1 n^{th} Order Derivative of Some Standard Functions

1. $y = (ax + b)^m$, where m is any real number.

Proof: $y = (ax + b)^m$

Differentiating w.r.t. x successively,

$$y_1 = ma\,(ax + b)^{m-1}$$
$$y_2 = m\,(m-1)\,a^2\,(ax + b)^{m-2}$$
$$y_3 = m\,(m-1)\,(m-2)\,a^3\,(ax + b)^{m-3},$$
$$\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$
$$\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$
$$y_n = m\,(m-1)\,(m-2)\ldots(m-n+1)\,a^n\,(ax + b)^{m-n}$$

Hence, $\dfrac{d^n}{dx^n}(ax+b)^m = m\,(m-1)\,(m-2)\ldots(m-n+1)\,a^n\,(ax+b)^{m-n}$

$$= \frac{m(m-1)\ldots(m-n+1)[(m-n)(m-n-1)\ldots3\cdot2\cdot1]a^n(ax+b)^{m-n}}{(m-n)(m-n-1)\ldots3\cdot2\cdot1}$$

$$= \frac{a^n m!(ax+b)^{m-n}}{(m-n)!}, \qquad \text{if } n < m$$

$$= n!\,a^n, \qquad \text{if } n = m$$

$$= 0, \qquad \text{if } n > m$$

2. $y = (ax + b)^{-m}$, where m is any positive integer.

Proof: $y = (ax + b)^{-m}$

Differentiating w.r.t. x successively,

$$y_1 = (-1)\,ma\,(ax + b)^{-m-1}$$
$$y_2 = (-1)^2\,m\,(m+1)\,a^2\,(ax + b)^{-m-2}$$
$$y_3 = (-1)^3\,m\,(m+1)\,(m+2)\,a^3\,(ax + b)^{-m-3}$$
$$\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$
$$\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$
$$y_n = (-1)^n\,m\,(m+1)\,(m+2)\ldots(m+n-1)\,a^n\,(ax + b)^{-m-n}$$
$$= (-1)^n \frac{(m+n-1)\ldots m(m-1)(m-2)\ldots2\cdot1}{(m-1)(m-2)\ldots2\cdot1}\frac{a^n}{(ax+b)^{m+n}}$$

Hence, $\dfrac{d^n}{dx^n}(ax+b)^{-m} = (-1)^n \dfrac{(m+n-1)!}{(m-1)!}\dfrac{a^n}{(ax+b)^{m+n}}$.

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Corollary 1: Putting $m = 1$, we get $\dfrac{d^n}{dx^n}(ax+b)^{-1} = (-1)^n n! \dfrac{a^n}{(ax+b)^{1+n}}$.

3. $y = \log(ax+b)$

Proof: $y = \log(ax+b)$

Differentiating w.r.t. x,

$$y_1 = \frac{a}{ax+b}$$

Differentiating $(n-1)$ times w.r.t. x,

$$\frac{d^{n-1}}{dx^{n-1}} y_1 = \frac{d^{n-1}}{dx^{n-1}}\left(\frac{a}{ax+b}\right)$$

$$\frac{d^{n-1}}{dx^{n-1}}\left(\frac{dy}{dx}\right) = \frac{a(-1)^{n-1}(n-1)!a^{n-1}}{(ax+b)^n}$$

Hence, $$\frac{d^n}{dx^n}\log(ax+b) = \frac{(-1)^{n-1}(n-1)!a^n}{(ax+b)^n}$$

4. $y = e^{ax}$

Proof: $y = e^{ax}$

Differentiating w.r.t. x successively,

$$y_1 = ae^{ax}$$
$$y_2 = a^2e^{ax}$$
$$y_3 = a^3e^{ax}$$
$$\ldots\ldots\ldots\ldots$$
$$\ldots\ldots\ldots\ldots$$
$$y_n = a^ne^{ax}$$

Hence, $$\frac{d^n}{dx^n}(e^{ax}) = a^ne^{ax}$$

5. $y = a^{mx}$

Proof: $y = a^{mx}$

Differentiating w.r.t. x successively,

$$y_1 = ma^{mx}\log a$$
$$y_2 = m^2a^{mx}(\log a)^2$$
$$y_3 = m^3a^{mx}(\log a)^3$$
$$\ldots\ldots\ldots\ldots$$
$$\ldots\ldots\ldots\ldots$$
$$y_n = m^na^{mx}(\log a)^n$$

Hence, $$\frac{d^n}{dx^n}(a^{mx}) = m^na^{mx}(\log a)^n$$

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6. $y = \sin(ax+b)$

Proof: $y = \sin(ax+b)$

Differentiating w.r.t. x successively,

$$y_1 = a\cos(ax+b) = a\sin\left(\frac{\pi}{2}+ax+b\right)$$

$$y_2 = a^2\cos\left(\frac{\pi}{2}+ax+b\right) = a^2\sin\left(\frac{2\pi}{2}+ax+b\right)$$

$$y_3 = a^3\cos\left(\frac{2\pi}{2}+ax+b\right) = a^3\sin\left(\frac{3\pi}{2}+ax+b\right)$$

.....

.....

$$y_n = a^n\sin\left(\frac{n\pi}{2}+ax+b\right)$$

Hence, $\dfrac{d^n}{dx^n}[\sin(ax+b)] = a^n\sin\left(\dfrac{n\pi}{2}+ax+b\right)$

Corollary 2: $\dfrac{d^n}{dx^n}[\cos(ax+b)] = a^n\cos\left(\dfrac{n\pi}{2}+ax+b\right)$

7. $y = e^{ax}\cos(bx+c)$

Proof: $y = e^{ax}\cos(bx+c)$

Differentiating w.r.t. x,

$$y_1 = ae^{ax}\cos(bx+c) + (-1)\,b\,e^{ax}\sin(bx+c)$$

$$y_1 = e^{ax}[a\cos(bx+c) - b\sin(bx+c)]$$

Let $a = r\cos\theta$, $b = r\sin\theta$

Then $r = \sqrt{a^2+b^2}$, $\theta = \tan^{-1}\dfrac{b}{a}$

$$y_1 = e^{ax}[r\cos\theta\cos(bx+c) - r\sin\theta\sin(bx+c)]$$

$$= re^{ax}\cos(bx+c+\theta)$$

Differentiating w.r.t. x,

$$y_2 = ae^{ax}r\cos(bx+c+\theta) - b\,e^{ax}r\sin(bx+c+\theta)$$

$$= re^{ax}[r\cos\theta\cos(bx+c+\theta) - r\sin\theta\sin(bx+c+\theta)]$$

$$= r^2e^{ax}\cos(bx+c+2\theta)$$

Differentiating n times w.r.t. x,

$$y_n = r^ne^{ax}\cos(bx+c+n\theta),$$

where $r = \sqrt{a^2+b^2}$, $\theta = \tan^{-1}\dfrac{b}{a}$.

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Hence, $\frac{d^n}{dx^n}[e^{ax}\cos(bx+c)] = r^n e^{ax}\cos(bx+c+n\theta)$,

where $r=\sqrt{a^2+b^2}, \theta=\tan^{-1}\frac{b}{a}$.

Corollary 3: $\frac{d^n}{dx^n}[e^{ax}\sin(bx+c)] = r^n e^{ax}\sin(bx+c+n\theta)$,

where $r=\sqrt{a^2+b^2}, \theta=\tan^{-1}\frac{b}{a}$.

Example 1: Find y_n if $y=\frac{x^n-1}{x-1}$.

Solution: $y=\frac{x^n-1}{x-1}=\frac{(x-1)(x^{n-1}+x^{n-2}+x^{n-3}+\ldots\ldots+1)}{(x-1)}$

$=x^{n-1}+x^{n-2}+x^{n-3}+\ldots\ldots+1$

Differentiating n times w.r.t. x,

$y_n=\frac{d^n}{dx^n}(x^{n-1}+x^{n-2}+x^{n-3}+\ldots\ldots+1)$

$=0$ $\left[\because \frac{d^n}{dx^n}(ax+b)^m=0, \text{ if } n>m\right]$

Example 2: Prove that $\frac{d^{2n}}{dx^{2n}}(x^2-1)^n=(2n)!$.

Solution: $(x^2-1)^n=(x^2)^n-{}^nC_1(x^2)^{n-1}+{}^nC_2(x^2)^{n-2}-\ldots\ldots(-1)^{n-1}$

$=x^{2n}-{}^nC_1x^{2n-2}+{}^nC_2x^{2n-4}-\ldots\ldots$

Differentiating $2n$ times w.r.t. x,

$\frac{d^{2n}}{dx^{2n}}(x^2-1)^n=\frac{d^{2n}}{dx^{2n}}(x^{2n}-{}^nC_1x^{2n-2}+{}^nC_2x^{2n-4}-\ldots\ldots)$

$=(2n)!$ $\left[\because \frac{d^n}{dx^n}(ax+b)^m = m!a^n, \text{ if } n=m;\ =0, \text{ if } n>m\right]$

Example 3: Find y_n, if $y=\frac{x}{(x+1)^4}$.

Solution: $y=\frac{x}{(x+1)^4}=\frac{(x+1)-1}{(x+1)^4}$

$=\frac{1}{(x+1)^3}-\frac{1}{(x+1)^4}$

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Differentiating n times w.r.t. x,

$$y_n = \frac{(-1)^n (n+2)!}{2!\,(x+1)^{n+3}} - \frac{(-1)^n (n+3)!}{3!\,(x+1)^{n+4}}$$

$$= \frac{(-1)^n (n+2)!}{2!\,(x+1)^{n+3}} \left[1 - \frac{n+3}{3(x+1)}\right]$$

$$= \frac{(-1)^n (n+2)!}{2!\,(x+1)^{n+3}} \left[\frac{3x-n}{3(x+1)}\right].$$

Example 4: Find y_n, if $y = \dfrac{x^2+4x+1}{x^3+2x^2-x-2}$.

Solution:

$$y = \frac{x^2+4x+1}{x^3+2x^2-x-2} = \frac{(x^2+4x+1)}{x^2(x+2)-(x+2)}$$

$$= \frac{x^2+4x+1}{(x+2)(x^2-1)} = \frac{x^2+4x+1}{(x+2)(x+1)(x-1)}$$

$$= \frac{A}{x+2} + \frac{B}{x+1} + \frac{C}{x-1}$$

[By partial fraction expansion]

$$(x^2+4x+1) = A(x+1)(x-1) + B(x+2)(x-1) + C(x+2)(x+1)$$

Putting $x = -2$, $A = -1$

Putting $x = -1$, $B = 1$

Putting $x = 1$, $C = 1$

$$y = \frac{-1}{x+2} + \frac{1}{x+1} + \frac{1}{x-1}$$

Differentiating n times w.r.t. x,

$$y_n = -\frac{(-1)^n n!}{(x+2)^{n+1}} + \frac{(-1)^n n!}{(x+1)^{n+1}} + \frac{(-1)^n n!}{(x-1)^{n+1}}$$

[Using Cor.1]

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Example 10: Find n^{th} order derivatives of

(i) $y = \sin 2x \sin 3x \cos 4x$ (ii) $y = \cos^4 x$

(iii) $y = \sin^5 x \cos^3 x$ (iv) $y = e^x (\sin x + \cos x)$

(v) $y = e^{x\cos\alpha} \cos(x \sin \alpha)$.

Solution:

(i) $$y = \sin 2x \sin 3x \cos 4x$$

$$= \frac{1}{2}(\cos x - \cos 5x)\cos 4x$$

$$= \frac{1}{2}(\cos x \cos 4x - \cos 5x \cos 4x)$$

$$= \frac{1}{4}(\cos 5x + \cos 3x - \cos 9x - \cos x)$$

Differentiating n times w.r.t. x,

$$y_n = \frac{1}{4}\left[5^n \cos\left(5x + \frac{n\pi}{2}\right) + 3^n \cos\left(3x + \frac{n\pi}{2}\right) - 9^n \cos\left(9x + \frac{n\pi}{2}\right) - \cos\left(x + \frac{n\pi}{2}\right)\right]$$ [Using Cor. 2]

(ii) $$y = \cos^4 x$$

$$= \left(\frac{2\cos^2 x}{2}\right)^2$$

$$= \frac{1}{4}(1 + \cos 2x)^2$$

$$= \frac{1}{4}(1 + \cos^2 2x + 2\cos 2x)$$

$$= \frac{1}{4}\left(1 + \frac{1 + \cos 4x}{2} + 2\cos 2x\right)$$

$$= \frac{1}{8}(3 + \cos 4x + 4\cos 2x)$$

Differentiating n times w.r.t. x,

$$y_n = \frac{1}{8}\left[4^n \cos\left(4x + \frac{n\pi}{2}\right) + 4.2^n \cos\left(2x + \frac{n\pi}{2}\right)\right]$$ [Using Cor. 2]

(iii)

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Example 15: Find the nth order derivative of $y = \cos^{-1}\left(\frac{x - x^{-1}}{x + x^{-1}}\right)$.

Solution: $y = \cos^{-1}\left(\frac{x - x^{-1}}{x + x^{-1}}\right) = \cos^{-1}\left(\frac{x^2 - 1}{x^2 + 1}\right)$

Putting $x = \tan\phi$,

$$y = \cos^{-1}\left(\frac{\tan^2\phi - 1}{\tan^2\phi + 1}\right) = \cos^{-1}(-\cos 2\phi)$$

$$= \cos^{-1}[\cos(\pi - 2\phi)] = \pi - 2\phi = \pi - 2\tan^{-1} x$$

Proceeding as in Example 13, we get

$$y = \cos^{-1}\left(\frac{x - x^{-1}}{x + x^{-1}}\right).$$

Example 16: If $y = x \log(1 + x)$, prove that $y_n = \frac{(-1)^{n-2}(n-2)!(x+n)}{(x+1)^n}$.

Solution: $y = x \log(1 + x)$

Differentiating w.r.t. x,

$$y_1 = \log(1+x) + \frac{x}{1+x} = \log(1+x) + 1 - \frac{1}{1+x}$$

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Example 18: If $y = x \coth^{-1} x$, prove that $y_n = \frac{(-1)^n (n-2)!}{2}\left[\frac{x+n}{(x+1)^n} - \frac{x-n}{(x-1)^n}\right], n \geq 2.$

Solution:

$$y = x \coth^{-1} x$$

$$= x \tanh^{-1}\left(\frac{1}{x}\right)$$

$$= x \cdot \frac{1}{2} \log \frac{1+\frac{1}{x}}{1-\frac{1}{x}}$$

$$= \frac{x}{2} \log\left(\frac{x+1}{x-1}\right)$$

$$y = \frac{1}{2}[x \log(x+1) - x \log(x-1)]$$

Proceeding as in Example 17, we get

$$y_n = \frac{(-1)^n (n-2)!}{2}\left[\frac{x+n}{(x+1)^n} - \frac{x-n}{(x-1)^n}\right], n \geq 2.$$

Example 19: If $y = (x-1)^n$, prove that $y + \frac{y_1}{1!} + \frac{y_2}{2!} + \frac{y_3}{3!} + \ldots + \frac{y_n}{n!} = x^n$.

Solution: $y = (x-1)^n$

Differentiating w.r.t. x successively,

$$y_1 = n(x-1)^{n-1}$$
$$y_2 = n(n-1)(x-1)^{n-2}$$
$$y_3 = n(n-1)(n-2)(x-1)^{n-3}$$
$$\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$
$$\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$
$$y_n = n!$$

Hence, $y + \frac{y_1}{1!} + \frac{y_2}{2!} + \frac{y_3}{3!} + \ldots + \frac{y_n}{n!}$

$$= (x-1)^n + \frac{n}{1!}(x-1)^{n-1} + \frac{n(n-1)}{2!}(x-1)^{n-2} + \frac{n(n-1)(n-2)}{3!}(x-1)^{n-3} + \ldots + \frac{n!}{n!}$$

$$= (x-1)^n + {}^nC_1(x-1)^{n-1} + {}^nC_2(x-1)^{n-2} + {}^nC_3(x-1)^{n-3} + \ldots + {}^nC_n$$

$$= [1+(x-1)]^n$$

$$= x^n.$$

[Using Binomial Expansion]

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